

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Ans:

- Season: Fall has the highest rentals, indicating favorable biking conditions.
- Month: June, September, and October show increased rentals due to pleasant weather and activities.
- Weekday: Peak rentals occur on Fridays and Sundays, reflecting leisure biking during weekends.
- Weather Situation: Clear weather significantly boosts rentals, while adverse conditions decrease demand.
- Holidays: Rentals increase during holidays as people engage in recreational activities.
- Additional Weather Features: Conditions like Light_snowrain and Misty negatively impact bike rental counts.

Overall, categorical variables such as season, month, weekday, and weather situation play a significant role in influencing bike rental demand. Understanding these relationships can help businesses strategize their marketing efforts, pricing, and availability during peak and off-peak times.

2. Why is it important to use drop_first=True during dummy variable creation?

Ans:

Using drop_first=True during dummy variable creation is important because it:

- Avoids Multicollinearity: Prevents redundancy by reducing correlation among dummy variables.
- Establishes a Reference Category: Allows for easier interpretation by designating one category as the baseline.
- Reduces Dimensionality: Streamlines the dataset by decreasing the number of dummy variables created.

This leads to a more interpretable and efficient model.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Ans:

Looking at the pair plot among the numerical variables, **temp** has the highest correlation with the target variable, **cnt**. This strong correlation suggests that as temperature increases, the number of bike rentals also tends to increase.

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

Ans:

Here are the assumptions of linear regression validated with one-liners and examples:

- Linearity: Checked using residual plots; for example, a scatter plot of predicted vs. actual values should show no discernible pattern.
- Independence: Validated with the Durbin-Watson test; a result close to 2 indicates no autocorrelation in residuals.
- Homoscedasticity: Assessed by plotting residuals against predicted values; a random scatter around zero confirms constant variance.
- Normality of Residuals: Evaluated using a histogram and Q-Q plot; a bell-shaped histogram and points along the reference line indicate normality.
- No Multicollinearity: Measured with Variance Inflation Factor (VIF); a VIF below 5 suggests multicollinearity is not an issue.
- Feature Significance: Assessed through p-values; coefficients with p-values less than 0.05 indicate that the features significantly contribute to the model.

The F-statistic is 248.4 with a p-value of 1.47e-186, indicating that the overall regression model is statistically significant, while the R-squared value of 0.833 and adjusted R-squared value of 0.829 suggest that approximately 83.3% of the variance in the dependent variable is explained by the independent variables.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Ans:

Based on the final model, the top three features contributing significantly to explaining the demand for shared bikes are **temp**, **winter**, and **sep**.

General Subjective Questions

1. Explain the linear regression algorithm in detail.

Ans:

Linear Regression Overview

- Concept: Linear regression models the relationship between a dependent variable Y and one or more independent variables $X_1, X_2, \dots, X_n$ using a linear equation:
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$
- Assumptions:
 - i. Linearity: Linear relationship between variables.
 - ii. Independence: Observations are independent.
 - iii. Homoscedasticity: Constant variance of residuals.
 - iv. Normality: Residuals are normally distributed.
 - v. No multicollinearity: Independent variables are not highly correlated.
- Cost Function: The Mean Squared Error (MSE) measures the average squared difference between actual and predicted values.
- Optimization: Coefficients are estimated using methods like Gradient Descent or the Normal Equation to minimize the cost function.

- Making Predictions: The trained model can predict new values by applying the learned coefficients to new data.
- Evaluation Metrics:
 - i. R-squared: Indicates the proportion of variance explained by the model.
 - ii. Adjusted R-squared: Adjusts R-squared for the number of predictors.
 - iii. F-statistic: Tests overall model significance.

2. Explain the Anscombe's quartet in detail.

Ans:

Anscombe's Quartet consists of four datasets that highlight the importance of visualizing data before analysis. Despite having nearly identical statistical properties, the datasets exhibit vastly different distributions when graphed.

Datasets Characteristics

Dataset	x Values	y Values	Mean of y	Variance of y	Correlation Coefficient	Linear Regression Equation
I	10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5	8.04, 6.58, 12.74, 7.74, 8.77, 9.26, 3.73, 1.78, 12.74, 4.69, 3.73	7.50	4.12	0.816	$y = 0.5x + 3$
II	8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8	6.58, 5.76, 7.71, 8.68, 6.56, 7.91, 6.89, 6.28, 5.64, 5.45, 5.16	6.58	0.12	0.0	$y = 7$
III	13, 8, 9, 11, 14, 6, 4, 12, 7, 5	7.74, 5.68, 8.65, 6.93, 8.10, 6.11, 2.74, 12.74, 4.31, 4.45, 4.21	6.58	4.12	0.816	$y = 0.5x + 3$
IV	8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8	12.74, 12.65, 12.50, 12.75, 12.85, 12.90, 12.91, 12.92, 12.93, 12.94, 12.95	12.74	0.12	-0.2	$y = -0.2x + 12.5$

- Statistical Similarity: All datasets share the same mean x , mean y , variance, and correlation coefficient, which can be misleading.

- Diverse Distributions: When graphed, each dataset reveals distinct relationships:
 - Dataset I: Strong linear relationship.
 - Dataset II: No relationship (constant y).
 - Dataset III: Strong linear relationship with outliers.
 - Dataset IV: Non-linear relationship.
- Emphasizes Visualization: Highlights the necessity of plotting data to uncover true relationships.
- Cautions Against Summary Statistics: Warns against relying solely on statistical measures.
- Educational Tool: Widely used in teaching statistical analysis and data exploration.

Anscombe's Quartet illustrates that different datasets can yield similar statistical summaries while possessing vastly different distributions and relationships. It underscores the critical importance of visualizing data in statistical practice.

3. What is Pearson's R?

Ans:

Pearson's R is a numerical summary that measures the strength and direction of the linear relationship between two variables.

- Positive Correlation: If the variables tend to increase or decrease together, the correlation coefficient will be positive ($r > 0$ or $0 < r < 1$). This means that as one variable increases, the other variable also tends to increase.
- Negative Correlation: If the variables move in opposite directions, with low values of one variable associated with high values of the other, the correlation coefficient will be negative ($r < 0$ or $-1 < r < 0$). In this case, as one variable increases, the other variable tends to decrease.
- No Correlation: A correlation coefficient of 0 indicates no linear association between the two variables.

Interpretation of Values:

- $r = +1$: Perfect positive correlation
- $0 < r < 1$: Positive correlation (the closer to 1, the stronger the relationship)
- $r = 0$: No correlation
- $-1 < r < 0$: Negative correlation (the closer to -1, the stronger the relationship)
- $r = -1$: Perfect negative correlation

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Ans:

What is Scaling?

Scaling refers to the process of transforming the values of features in a dataset to a common scale without distorting differences in the ranges of values. This is particularly important in machine learning and statistical analysis, as many algorithms perform better or converge faster when the data features are on a similar scale.

Why is Scaling Performed?

- Improved Model Performance
- Avoid Feature Dominance
- Enhanced Interpretability
- Required by Certain Algorithms

Difference Between Normalized Scaling and Standardized Scaling

Normalized Scaling (Min-Max Scaling):

- Scales the data to a specific range, typically $[0, 1]$.
- $X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$
- Best suited for datasets that do not follow a Gaussian distribution and when you want to preserve the relationships between the original values.
- All transformed values fall within the range of 0 to 1.

Standardized Scaling (Z-score Scaling):

- Centers the data around the mean and scales it based on the standard deviation.
- $X' = \frac{X - \mu}{\sigma}$ where μ is the mean and σ is the standard deviation of the feature.
- Preferred when the data follows a Gaussian distribution, as it produces a distribution with a mean of 0 and a standard deviation of 1.

- Transformed values can take any real number, focusing on how many standard deviations away they are from the mean.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

Ans:

Infinite VIF indicates high correlation among predictors, leading to instability in the regression model and unreliable coefficient estimates. To resolve this, it's essential to identify and remove redundant or perfectly correlated variables.

- Perfect Multicollinearity
- Redundant Variables
- Insufficient Data
- Data Error

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

Ans:

What is a Q-Q Plot?

A Q-Q (Quantile-Quantile) plot is a graphical tool used to compare the quantiles of a dataset against the quantiles of a theoretical distribution (often a normal distribution). It helps assess whether the data follows a specified distribution. The points on the plot represent the quantiles of the data on the y-axis and the quantiles of the theoretical distribution on the x-axis.

Use of Q-Q Plot

- Assessing Normality: The primary use of a Q-Q plot is to visually evaluate whether the residuals of a regression model are normally distributed, which is a key assumption of linear regression.
- Identifying Deviations: It helps identify deviations from normality, such as skewness or heavy tails, which can impact the validity of statistical tests and model interpretations.

- Comparing Distributions: Q-Q plots can be used to compare two datasets to see if they come from the same distribution, making it useful in various statistical analyses beyond linear regression.

Importance of Q-Q Plot in Linear Regression

- Validation of Assumptions: Normality of residuals is one of the key assumptions in linear regression. A Q-Q plot allows you to check this visually, ensuring the reliability of hypothesis tests and confidence intervals derived from the model.
- Model Diagnostics: By examining the Q-Q plot, you can detect potential issues with the model, such as outliers or non-normality, prompting further investigation or model adjustments.
- Impact on Inference: If residuals are not normally distributed, it can affect the accuracy of the estimated coefficients and the validity of statistical inferences made from the model.

Interpretation of a Q-Q Plot

- Points on the Line: If the points fall approximately along the diagonal line, it indicates that the data follows the theoretical distribution closely (e.g., normal distribution).
- Points Curving Upwards or Downwards: Deviations from the line (e.g., points curving upwards at the ends) suggest that the data may have heavier tails than the normal distribution (indicating potential outliers).
- S-shaped Pattern: This pattern suggests that the data may be skewed.